REFINING A SYNTOMIC, SMOOTH OR ETALE COVERING OF A CLOSED SUBSCHEME TO A COVERING OF THE AMBIENT SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the local data, the covering of the image and the inclusion of the complement are all inside the counted span.

Editorial note. The label lemma-prepare-closed-immersion-almost-cocontinuous occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Syntomic, smooth and etale coverings of a scheme. Same chapter, the topology generated by syntomic, smooth respectively etale morphisms on the category of schemes over a base. Supplies the notion of covering refined in the statement and verified in the proof.
- Covers of a closed subscheme lifted to the ambient scheme. Same chapter, the result producing, for each member of a covering of X, morphisms to Y whose base change to X refines the given covering member. Supplies the families V_{ij} and the morphisms V_{ij} x_Y X -> U_i.
- Open immersion of the complement of a closed subscheme. Same chapter, the open immersion Y minus f(X) -> Y used to cover the points outside the image. Supplies the additional member of the covering that handles the complement of f(X).

Source reference. Original lemma 04E3: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6857–6857. Same-chapter navigation pointer invoked at proof lines 6897-6898, where the author produces morphisms from Y covering the closed subscheme. The pointer only resolves that reference for the reader; the cited lemma is not reproduced inside the counted span, and no bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to Y$ be a closed immersion of schemes. Let $\{U_i \to X\}$ be a syntomic (resp. smooth, resp. étale) covering. There exists a syntomic (resp. smooth, resp. étale) covering $\{V_j \to Y\}$ such that for each j, either $V_j \times_Y X = \emptyset$, or the morphism $V_j \times_Y X \to X$ factors through U_i for some i.*

Proof. For each i we can choose syntomic (resp. smooth, resp. étale) morphisms $g_{ij} : V_{ij} \to Y$ and morphisms $V_{ij} \times_Y X \to U_i$ over X, such that $\{V_{ij} \times_Y X \to U_i\}$ are Zariski coverings, see Lemma 04E3. This in particular implies that $\bigcup_{ij} g_{ij}(V_{ij})$ contains the closed subset $f(X)$. Hence the family of syntomic (resp. smooth, resp. étale) maps g_{ij} together with the open immersion $Y \setminus f(X) \to Y$ forms the desired syntomic (resp. smooth, resp. étale) covering of Y. $\square$